

STARLAND RAT PYTHON LOADER MASQUERADING

pythonw.exe executes a compiled loader
disguised as LICENSE.txt.

HUNTING**CONFIDENCE: HIGH**

STARLAND RAT

Hunt for
pythonw.exe
receiving
the source-
observed
text-file
loader
name as a
command-
line
argument.

TROJANIZED INSTALLER

BUNDLED PYTHONW

LICENSE.TXT LOADER

IN-MEMORY DECRYPT

STARLAND RAT

REPORT DATE**22 July 2026****VERSION****v2.0****PRIMARY TELEMETRY****Microsoft Defender XDR ·
DeviceProcessEvents****ATT&CK / SCOPE****T1059.006 · T1036.008**

Independent defensive research. Source facts, analytical assessment, assumptions and detection logic are intentionally separated. Every query requires environment-specific validation.

01

EXECUTIVE ASSESSMENT

Decision-grade summary for threat research, detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Starland RAT Python Loader Masquerading

Cisco Talos analyzed trojanized installers bundling pythonw.exe with a compiled Python loader named LICENSE.txt. The runtime-plus-masqueraded-filename intersection is a compact campaign pivot.

WHY IT MATTERS

Operational detection value

A text-extension argument to pythonw.exe may be low-volume and directly supported by the source, but the filename is attacker-controlled and trivially changed.

High 01

ANALYTICAL CONFIDENCE



Focused 02

TELEMETRY COVERAGE



Hunting 03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACT

Documented behavior

The analyzed NSIS installer executed LICENSE.txt through pythonw.exe before in-memory Starland RAT execution.

ANALYTICAL ASSESSMENT

Interpretation

The process-filename pair is suitable for campaign hunting; it is not durable enough for a universal production rule.

DETECTION ASSUMPTION

Required condition

The loader filename remains visible in ProcessCommandLine and is not hidden behind a wrapper.

Evidence boundary

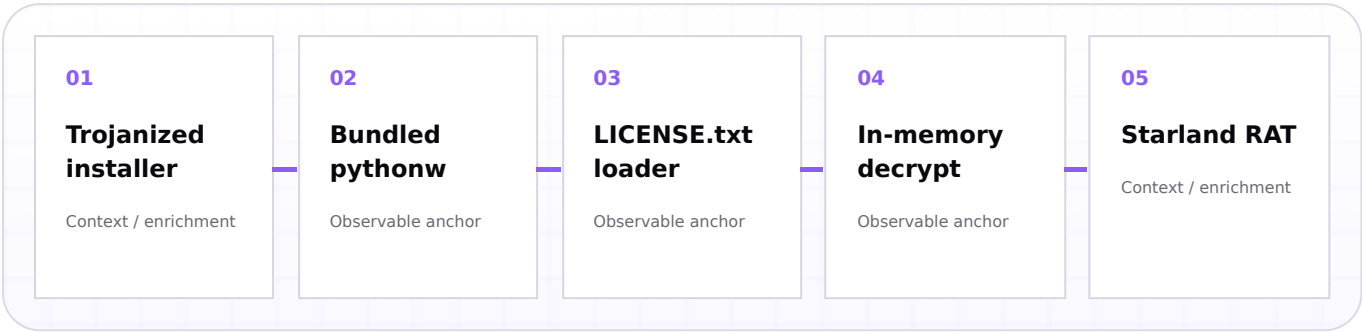
It does not prove Python bytecode content, decryption, Starland execution or ClickFix initial access.

02

THREAT MECHANICS

Behavioral decomposition and observable evidence boundaries.

ATTACK FLOW



STAGE-BY-STAGE DETECTION VALUE

STAGE	BEHAVIOR	EVIDENCE EXPECTATION	USE
01	Trojanized installer	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
02	Bundled pythonw	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
03	LICENSE.txt loader	The analyzed NSIS installer executed LICENSE.txt through pythonw.exe before in-memory Starland RAT execution.	Primary anchor
04	In-memory decrypt	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
05	Starland RAT	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment

Observable core

A match indicates the exact runtime and masqueraded filename relationship described by the source.

Non-observable boundary

It does not prove Python bytecode content, decryption, Starland execution or ClickFix initial access.

TELEMETRY & EVIDENCE MODEL

Minimum data contract required for a defensible implementation.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR · DeviceProcessEvents

The primary platform is selected because it observes the strongest source-supported behavior with the most direct event relationship.

EVIDENCE BOUNDARY

A match is an investigative lead.

It does not prove Python bytecode content, decryption, Starland execution or ClickFix initial access.

REQUIRED FIELD CONTRACT

TIER	DATA	PURPOSE	RELIABILITY
REQUIRED	Timestamp, DeviceId, DeviceName, FileName, FolderPath, ProcessCommandLine, InitiatingProcessFileName, AccountName, SHA1, ReportId	Core filtering, grouping and investigation output.	Documented schema; population must be verified.
OPTIONAL	Signer, hash, prevalence, file origin, software inventory, device group.	Benign separation and prioritization.	Environment-dependent.
ENRICHMENT	Threat intelligence, change windows, asset criticality and identity context.	Disposition and escalation.	Integration-dependent.

Completeness

Confirm table coverage, retention, field population and event latency before measuring prevalence.

Normalization

Validate case, path, quoting, URI encoding and command-line preservation against raw events.

Visibility gap

Renaming the loader or using a wrapper evades this campaign-specific hunt.

04

DETECTION LOGIC

Source-grounded implementation with explicit assumptions and outcome boundaries.

HUNTING

DETECTION OBJECTIVE

Hunt for pythonw.exe receiving the source-observed text-file loader name as a command-line argument.

IMPLEMENTATION STATUS

Hunting - Untested implementation sketch

Syntax and schema are documented; local execution, field population and false-positive performance are not established.

KQL · DEMONSTRATIVE

```
// Title: Starland RAT Python Loader Masquerading as LICENSE.txt
// Status: Hunting query - campaign-specific artifact
// Confidence: High
// Platform: Microsoft Defender XDR / Microsoft Sentinel with MDE tables
// Telemetry: DeviceProcessEvents
// Purpose: Hunt for pythonw.exe executing the source-observed compiled Python
// loader masquerading with a text-file name.
// ATT&CK: T1059.006, T1036.008
// Limitation: The filename is mutable and may be hidden by an execution wrapper.

DeviceProcessEvents
| where FileName =~ "pythonw.exe"
| where ProcessCommandLine contains "LICENSE.txt"
| project Timestamp, DeviceId, DeviceName, AccountName,
    FolderPath, ProcessCommandLine,
    InitiatingProcessFileName,
    InitiatingProcessCommandLine,
    SHA1, ReportId
```

What a match means

A match indicates the exact runtime and masqueraded filename relationship described by the source.

What a match does not prove

It does not prove Python bytecode content, decryption, Starland execution or ClickFix initial access.

TRIAGE & INVESTIGATION

Evidence-led workflow for defensible disposition.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the match.
Review the raw event, timestamp, entity relationship and exact matching fields.
- 02

Establish legitimacy.
Identify the software or change owner, expected path, signer and authorized maintenance window.
- 03

Reconstruct origin.
Pivot to process, file, identity, network or application events before and after the match.
- 04

Measure prevalence.
Compare hashes, paths, commands, users and devices across the retained fleet.
- 05

Document confidence.
Separate confirmed evidence, inference, assumptions and unresolved gaps.

DECISION MODEL

Approved context → tune narrowly

Known owner, expected software path and repeatable behavior can support a documented exclusion.

Unknown context → deepen review

Rare or risky context without corroboration remains suspicious or insufficient evidence.

Corroborated activity → escalate

Malicious follow-on process, file, network, persistence or identity evidence supports incident escalation.

Host

Coverage, exposure, recent changes and neighboring alerts.

User

Session, intent, role and authorization context.

Process

Parent, child, path, command, signer and original filename.

File

Hash, signature, origin, prevalence and creation timeline.

Network

DNS, proxy, TLS and destinations following the event.

Identity

Sign-ins, token use, privilege and resource access.

OUTCOME	EVIDENCE PROFILE	ACTION
Benign	Approved and repeatable behavior with strong ownership.	Document and tune narrowly.
Suspicious	Risky context without decisive malicious corroboration.	Deepen endpoint or application review.
Likely malicious	Source-compatible match plus corroborated malicious activity.	Escalate incident response.
Insufficient evidence	Missing telemetry or unresolved ownership.	Retain evidence and validate gaps.

06

FALSE POSITIVES & VALIDATION

Operational controls required before alerting.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	LIKELIHOOD	TUNING ACTION
Custom Python packaging	High	Review owner, path, signer, prevalence and change context before exclusion.
Developer experiments	High	Review owner, path, signer, prevalence and change context before exclusion.
Installer test harnesses	Med	Review owner, path, signer, prevalence and change context before exclusion.
Security simulations	Med	Review owner, path, signer, prevalence and change context before exclusion.

Baseline plan

- Run retrospectively over a representative retention window.
- Measure volume by device, user, process, path and software owner.
- Inspect recurring legitimate patterns before adding exclusions.
- Preserve raw examples for peer review.

Validation procedure

- Confirm every required field against raw events.
- Execute only safe, authorized lab or historical validation.
- Compare query output with endpoint or application ground truth.
- Record success, rejection and sensor-failure criteria.

PROMOTION GATE

Hunting

Source-grounded hypothesis.

Candidate

Schema and baseline complete.

Production

Measured precision and runbook.

Optimized

Stable feedback loop.

Promotion criteria

Known alert volume, documented exclusions, representative validation, triage ownership, data-quality monitoring and measurable precision.

Failure condition

Renaming the loader or using a wrapper evades this campaign-specific hunt.

07

ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

COVERAGE MAPPING

MAPPING	COVERAGE STATEMENT	BOUNDARY
T1059.006 · T1036.008	Partial behavioral or campaign-artifact coverage of the source-supported observable.	It does not prove Python bytecode content, decryption, Starland execution or ClickFix initial access.

ANALYST CONCLUSION

Assessment, not certainty

The pairing is assessed as a high-confidence hunting lead, with promotion dependent on local prevalence and file-content validation.

Residual risk

Renaming the loader or using a wrapper evades this campaign-specific hunt.

REFERENCES

- 1. UAT-11795 deploys novel Starland RAT and bespoke WLDR C2 implant in financially motivated campaign: [https://blog\[.\]talosintelligence\[.\]com/uat-11795-deploys-novel-starland-rat-and-bespoke-wldr-c2-implant-in-financially-motivated-campaign/](https://blog[.]talosintelligence[.]com/uat-11795-deploys-novel-starland-rat-and-bespoke-wldr-c2-implant-in-financially-motivated-campaign/)
- 2. DeviceProcessEvents table in the advanced hunting schema: [https://learn\[.\]microsoft\[.\]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table](https://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table)

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v2.0	22 July 2026	Hunting	Detection Engineering

Change summary

Migrated to the permanent Threat Research Editorial HTML master; technical query and evidence classification preserved.

Publication control

No customer data, credentials or environment-specific internal indicators are included. URLs are defanged in the publication artefact.